Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CSDE731]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

ARTIFICIAL NEURAL NETWORKS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	List the different models of artificial neurons?	**04**
	b.	Describe the 3 major categories of Artificial Neural Networks.	**06**
	c.	What types of learning can neural networks perform?	**04**
2.	**a.**	Describe the Back propagation algorithm in Neural Networks.	**07**
	b.	Discuss the concept behind the Credit Assignment Problem.	**07**
3.	**a.**	Discuss the concept of Associative memory in Neural Networks.	**07**
	b.	Distinguish between ART1 and ART2 along with its applications.	**07**
		OR	
4.	**a.**	What is associative memory giving its types in neural network?	**07**
	b.	Explain Kohonen Self Organizing feature map.	**07**
5.	**a.**	Discuss the different types of Regularization techniques.	**10**
	b.	Write a note on Deep feedforward networks.	**04**
		OR	
6.	**a.**	What is dropout in deep learning explain with an example?	**04**
	b.	What is initialization in Neural Network?	**06**
	c.	Describe Gradient descent strategies in deep learning.	**04**
7.	**a.**	Write a note on Content based image Retrieval.	**06**
	b.	Describe Convolution Neural Networks with an example.	**08**
		OR	
8.	**a.**	Discuss the applications of Convolution Neural Network.	**07**
	b.	Write a note on VGGNet, GoogLeNet.	**07**